

Analysis of Crashes Involving Micromobility Devices

Enock Mwambeleko¹ , Priyanka Alluri¹, Thobias Sando², HM Nayem¹, Amy Aida¹ , and Janeroze Matyenyi²

Abstract

Shared micromobility systems have become increasingly popular for short-distance travel. Different cities have been adopting micromobility systems to mitigate several transportation-related problems. Micromobility devices include e-scooters and e-bikes. However, the safety impacts of these devices have not been extensively studied. This study investigates the safety implications of various on-street facilities for micromobility users. It estimates crash modification factors for micromobility crashes in Florida for roadway segments and signalized intersections. This research utilizes a negative binomial Lindley model to analyze crash data involving micromobility users from 2021 to 2023, identifying significant factors contributing to micromobility crashes. For roadway segments, the presence of on-street parking and sharrow lanes were found to increase the likelihood of micromobility crashes. In contrast, the presence of sidewalk barriers was associated with a reduction in such crashes. At signalized intersections, the presence of multiple approaches with exclusive left-turn lanes, exclusive right-turn lanes, and permitted left-turn signals were associated with an increased probability of micromobility crashes. Conversely, intersections with multiple approaches featuring protected-permitted left-turn signals were associated with a reduction in micromobility crashes. The findings of this study provide valuable insights for transportation agencies planning to incorporate micromobility systems in their jurisdictions, helping to improve the overall safety of these systems.

Keywords

micromobility, crash modification factors, negative binomial Lindley model

Shared micromobility systems have rapidly emerged as a popular transportation option for short distances, with 133 million trips made in the United States alone in 2023, a 16% increase from 2022 (1). These systems have shown significant potential in alleviating traffic congestion by reducing the number of vehicle trips, facilitating short-distance travel, extending the reach to public transportation, and addressing first- and last-mile travel challenges (2, 3). This potential to alleviate traffic congestion is a promising aspect of shared micromobility systems in urban areas. Additionally, by decreasing the number of vehicle trips in urban and suburban areas, micromobility devices serve as an effective tool for decarbonizing transportation, reducing noise pollution, and enhancing the overall livability and health of cities (4). The adoption of micromobility also promotes active mobility, which is associated with numerous health benefits for users (5).

Micromobility devices are motorized transportation devices that are mainly designed not to exceed 20 mph

on level ground (6). These devices can be privately owned and operated or used as part of a shared system. Examples of micromobility devices include electric bikes (e-bikes), electric scooters (e-scooters), shared bicycles, and electric pedal-assisted bicycles. Shared micromobility devices are generally categorized into two main types: docked (station-based) systems and dockless systems. In station-based systems, users start and end their trips at designated locations (stations), whereas dockless systems allow users to start and end their trip almost anywhere within the city, offering greater flexibility and convenience (4).

¹Department of Civil & Environmental Engineering, Florida International University, Miami, FL

²School of Engineering, University of North Florida, Jacksonville, FL

Corresponding Author:

Enock Mwambeleko, emwam001@fiu.edu

The spread of micromobility use has brought about a rapid increase in micromobility crashes. In the United States, the number of crashes involving e-scooters increased from 8,016 in 2017 to 14,641 in 2018 (7). This may be a result of the rapid adoption of these devices, which left planners and engineers with little time to assess the impacts on existing infrastructure and the necessary safety measures to meet user needs (8). Micromobility users, like motorcyclists, are not physically protected when riding, making them particularly vulnerable to serious injuries, especially on collision with motor vehicles. Additionally, the high center of gravity and smaller wheels on e-scooters make them less stable, increasing the likelihood of severe outcomes during crashes (9). In recent years, transportation agencies have implemented diverse strategies to combat the occurrence and severity of micromobility crashes, such as providing bike facilities (e.g., installing bike lanes, bicycle boulevards), proper parking corrals, and reducing speed limits. However, the specific impact of such strategies on reducing micromobility-related crashes remains uncertain, highlighting the need for further research in this area.

This study aims to assess the safety implications of various on-street facilities on micromobility users by analyzing crash data and evaluating the effectiveness of different infrastructure elements. The study will utilize statistical models to identify significant factors contributing to micromobility-related crashes and estimate crash modification factors (CMFs) to quantify the impact of each variable. The findings of this study will provide valuable insights to urban planners and transportation engineers, assisting them in designing safer streets and implementing effective strategies to reduce the risk of crashes involving micromobility users.

Literature Review

Crash Frequency Modeling

CMFs measure the potential change in the number of crashes observed at a particular location when a specific treatment or countermeasure is applied (10). A crash reduction factor (CRF) provides an estimate of the percentage of crashes that are reduced when a safety countermeasure is applied (11). A mathematical relationship between CMF and CRF is presented in Equation 1.

$$\text{CRF} = 100(1 - \text{CMF}) \quad (1)$$

CMFs are typically estimated using two primary methods: observational before–after studies and cross-sectional studies. Cross-sectional studies are often preferred because isolating the impact of a specific countermeasure from other influencing factors is difficult in before–after studies (12). Moreover, before–after studies

are impractical when there is insufficient crash data for the period before and after the implementation of treatment or when the installation date of the treatment is unknown. In such cases, cross-sectional studies are more useful (10).

In cross-sectional studies, the safety performance of a site with a specific treatment is compared with a similar site without that treatment (12). Safety performance functions (SPFs) or crash prediction models (CPMs) are used for the safety analysis. These models relate crash frequency to the traffic and geometric characteristics of the site (13). The most commonly used probabilistic structures for developing CPMs are the Poisson and negative binomial (NB) (or Poisson-gamma) regression models (14). Of the two, the NB model is preferred because of its ability to handle overdispersion in crash data. However, the NB model has limitations when dealing with datasets that contain many zero-crash observations (14). These zero-crash observations occur because road crashes are rare and random, resulting in many locations with no observed crashes. Additionally, vehicle crashes are often underreported, or crash details may be omitted during reporting, contributing to the excess of zeros in the data.

To address this limitation, zero-inflated models, such as the zero-inflated Poisson and zero-inflated negative binomial (ZINB), have been proposed for modeling traffic crashes (14–16). These models assume a dual-state data-generating process to account for the excess zeros: an “inherently safe” or zero-crash state and a nonzero state, which may happen to record zero crashes during the analysis period (17). Although zero-inflated models have shown strong performance in handling crash data with excess zeros (18–20), criticisms of their underlying assumptions (14, 17, 21) have led researchers to explore alternative approaches for modeling zero-inflated data such as mixed modeling techniques. Mixed models use combined probability distributions to account for overdispersion in crash data. One such model is the NB-Lindley (NB-L) model, which is a mixture of the NB and Lindley distributions (22). This model has a higher probability mass at zero and a heavier tail than the traditional NB model, providing better statistical fit for overdispersed count data with an excess of zero outcomes (22–25).

The NB-L model was first applied to crash count data analysis by Lord and Geedipally. Their study demonstrated the model’s effectiveness in handling crash datasets characterized by excess zeros and a heavy tail while still preserving the properties of the NB model (24). Similar findings were reported by Shaon and Qin who applied the NB-L model to evaluate the safety impacts of lane and shoulder widths on rural highways (22). In a study aimed at developing SPFs for nonfederal aid

system roads in Virginia, a comparison of the NB-L model with other forms of the NB model showed that the NB-L model provided a better statistical fit for datasets with many zero-crash observations (26). More recently, an assessment of the performance of the NB-L model in hotspot identification showed a high probability of correctly identifying nonhazardous locations (27).

Several extensions of the NB-L model have been developed and studied in the literature. For example, Geedipally et al. (28) developed an NB-L generalized linear model (NB-L GLM) for crash data analysis. The model was evaluated using observed and simulated crash data and was found to outperform traditional NB and ZINB models when dealing with highly dispersed, skewed, or zero-inflated data. Anis et al. used a random parameters NB-L (RPNB-L) model to investigate the factors contributing to pedestrian crashes near bus stops (29). The model addressed the challenges of unobserved heterogeneity and excess zero-crash observations (29–31). Another study developed and evaluated the performance of an NB-L model with time-dependent parameters (32). The model provided superior goodness of fit compared with the NB and NB-L models. Other variations of the NB-L model that have shown superior performance when compared with the NB model include the correlated RPNB-L (33), the finite-mixture NB-L (34), and the grouped RPNB-L (35).

Factors Influencing Micromobility Crashes

The built environment significantly influences micromobility crashes. Research shows that high population density and mixed land-use areas are significantly correlated with increased crash frequency. Conversely, the presence of dedicated bike lanes and protected cycle tracks markedly reduces collision rates, underscoring the importance of complete streets in enhancing micromobility safety (7). A study on e-scooter crashes in Austin, TX, found that a higher percentage of sidewalks was associated with a reduction in the number of e-scooter injury crashes. Additionally, census tracts with a high proportion of public transport users had fewer e-scooter injury crashes (36). Crash-prone areas typically include roadway intersections and sidewalks, where the interaction between pedestrians, micromobility users, and motor vehicles is usually fatal (37).

Research on micromobility user safety perceptions consistently highlights the importance of dedicated road infrastructure. One study found that shared bike users reported feeling safest when riding on dedicated cycling infrastructure (38). Similarly, another study found that micromobility users experience a greater sense of safety and comfort when using pathways separated from parked or moving vehicles, such as protected bike lanes

(39). Furthermore, research shows that most micromobility users perceive streets with clearly delineated spaces for different modes of transport and a high density of designated crosswalks as safer (40).

Several studies have highlighted factors associated with injury severity and risk of micromobility crashes. Human behaviors such as not wearing a helmet and riding under the influence of alcohol further exacerbate the risk of injuries (41). Yang et al. found that speeding and micromobility usage at night when there are poor lighting conditions are highly associated with fatal crashes (42). Collision type also affects the injury severity of micromobility crashes, with collisions with vehicles and falling off linked to higher injury severity.

The objective of this study is to develop CMFs for micromobility crashes using cross-sectional analysis. The relevant multivariate regression models were developed using an NB-L model.

Data

This study focused on road segments and signalized intersections in the City of Miami Beach and Downtown Miami regions in Florida. Geometric, traffic, and crash data were obtained from multiple sources, including Google Maps, Florida geographic information system (GIS) shapefiles, Signal Four Analytics, and Florida Department of Transportation's (FDOT) Roadway Characteristics Inventory (RCI). A detailed description of the data collection efforts and variables included in the study are presented in the following subsections.

Roadway Segment Data

The road geometry data for segments were manually collected using Google Maps. The following 10 variables of interest were considered:

- Section annual average daily traffic (AADT): the estimated average number of vehicles that travel on the road segment daily over the course of a year. These values were obtained from FDOT ArcGIS shapefiles;
- Sidewalk barrier: a physical barrier that separates vulnerable road users using a sidewalk from regular traffic lanes. Examples include parking areas, physical traffic barriers, guardrails, and trees;
- Number of lanes;
- Bike lane;
- Sharrow lanes;
- Median type;
- Sidewalk;
- Trafficway/road type; and
- On-street parking.

Table 1. Descriptive Statistics of Roadway Segments

Attribute	Attribute category		Total segment length (mi)	Frequency
Number of lanes	1 and 2		23.2	22
	3 and 4		57.7	44
	5 and 6		14.4	12
Median type (mi)	No median		40.9	34
	Paved		17.3	14
	Curb and vegetation		37.1	30
Bike lane (mi)	Absent		63.5	52
	Present		31.8	26
On-street parking (mi)	No		30.1	28
	One side		11.8	10
	Both sides		53.4	40
Sidewalk (mi)	Absent		15.7	14
	Present		79.5	64
Sidewalk barrier (mi)	Absent		48.6	40
	Present		46.6	38
Trafficway (mi)	Undivided		58.2	48
	Divided		37.1	30
Sharrow lanes (mi)	Absent		43.8	38
	Present		51.4	40
Continuous variables	Mean	SD	Minimum	Maximum
Crash frequency (2021–2023) ^a (crashes/year/mi)	0.4	0.9	0	4
Section AADT (vpd)	17,574	13,615	1,900	53,000
Roadway length (mi)	1.0	0.7	0.1	2.9

Note: AADT = annual average daily traffic; SD = standard deviation; vpd = vehicles per day.

^aCrashes involving micromobility users.

Table 1 presents the descriptive statistics of the roadway segment data. The classification and categorization of the variables were in alignment with the guidelines provided in the FDOT RCI handbook (43).

Signalized Intersection Data

Spatial analysis in ArcGIS was conducted to identify the signalized intersections in the City of Miami Beach and Downtown Miami (Figure 1). The identification process followed several key steps. First, spatial analysis using ArcGIS was employed. From the RCI data, the study utilized two shapefiles: the “signals” shapefile, which contains signalized intersections, and the “intersections” shapefile, which contains both signalized and unsignalized intersections.

The next step involved using the spatial join feature to determine which intersections were signalized by matching the data from the two shapefiles. However, challenges arose with the existing GIS shapefiles, including multiple point features at the same intersection and point features incorrectly located on roadway segments.

To address these issues and ensure accuracy in identifying all signalized intersections, the study opted to manually create point features using Google Earth maps in conjunction with Google Street View. These manually

created point features were saved as .kmz files and then converted to shapefiles in ArcMap. This manual verification and creation process was necessary to ensure the precision and reliability of the data for signalized intersections in the Downtown Miami and the City of Miami Beach areas.

The analysis considered the following 10 variables (the data for these variables were manually collected using Google Maps and Google Earth):

- Major road AADT: the estimated AADT on the major road approaching the intersection. This information was retrieved from the FDOT ArcGIS shapefiles;
- Number of intersection legs;
- Number of approaches with exclusive right-turn lane;
- Number of approaches with exclusive left-turn lane;
- Presence of a bike facility;
- Presence of lighting;
- Number of approaches with permitted left-turn signal;
- Number of approaches with protected-permitted left-turn signal; and
- Number of approaches with no turn on red sign.

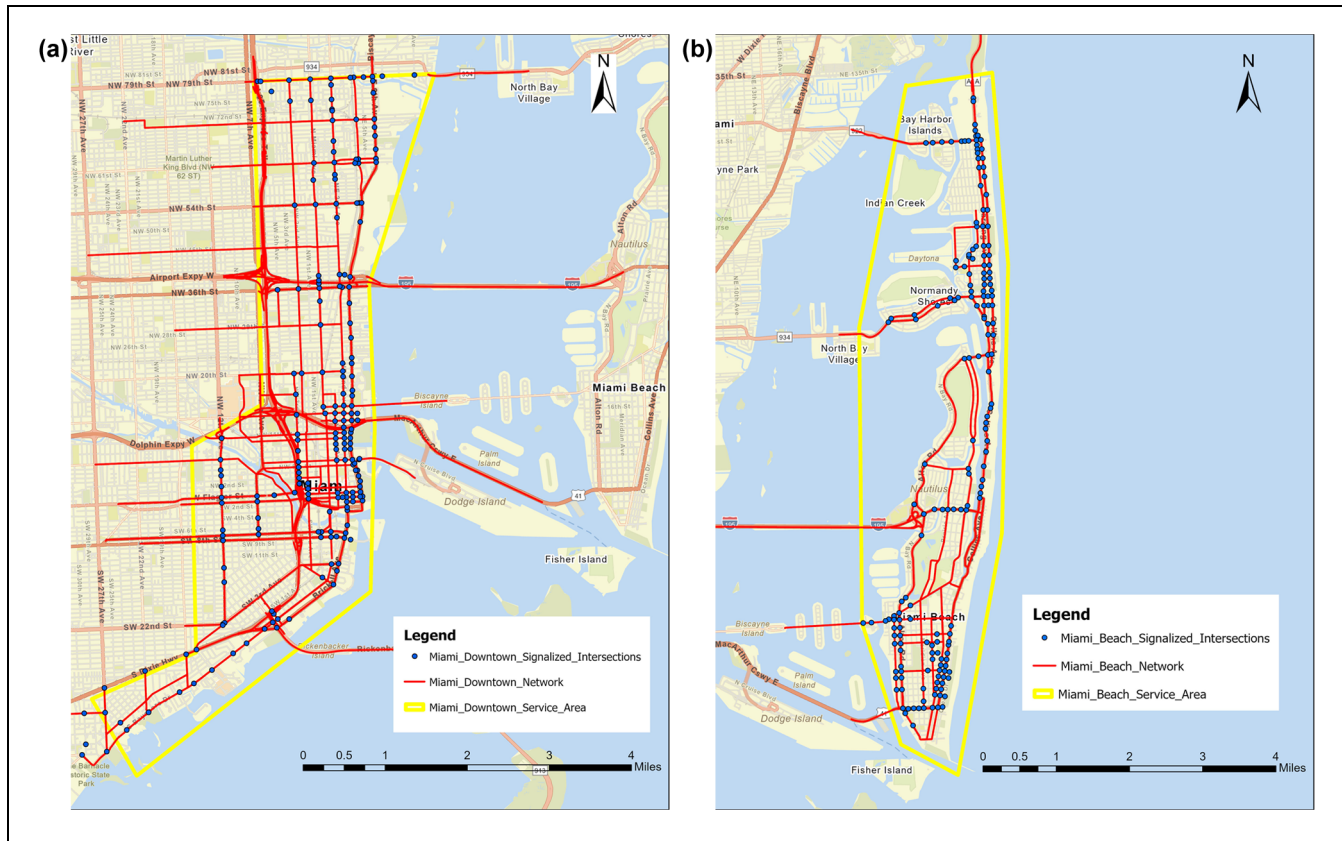


Figure 1. Signalized intersections in (a) Downtown Miami and (b) City of Miami Beach.

Table 2 summarizes the descriptive statistics for the signalized intersections. For each variable, a specific base condition was considered to help understand the relative effect of different traffic control measures and physical characteristics on the safety performance of signalized intersections.

Crash Data

Data from crashes involving micromobility devices that occurred in Florida between January 2021 and December 2023 were obtained from the Signal Four Analytics database. Signal Four Analytics is an interactive, web-based system designed to support law enforcement and research institutions' crash mapping and analysis needs. The process of obtaining the crash data involved three steps. First, crash reports for the City of Miami Beach and Downtown Miami regions were extracted from the Signal Four Analytics database. Since no specific variable in the crash database could be used to identify micromobility-related crashes, a text-mining algorithm was developed to identify crash reports involving shared micromobility devices. Crash reports with the keywords "micromobility," "scooter," "e-scooter," "motorized," "e-bike," "Citi Bike," "DECOBIKE," and "electric" were

extracted using the Tesseract package in R software. This step was important to reduce the number of crash reports that needed to be manually reviewed to verify that the crashes indeed involved micromobility devices.

After the text-mining process, the next step was to manually review the crash reports that contained at least one of the keywords to confirm they actually involved micromobility devices. This step was important since crash reports could contain the keywords but not involve micromobility. For example, a crash that involves an electric wheelchair would be included as it has the keyword "electric." Such crashes were removed from the dataset. A total of 301 crashes involving shared micromobility devices were identified. Of these, 156 (52%) occurred in Downtown Miami, and 155 (48%) occurred in the City of Miami Beach. Figure 2 displays the summary of the descriptive characteristics of the crash data. For the purpose of this analysis, intersection-related crashes were defined based on the 250-ft buffer from the intersection. Figure 3 presents the spatial distribution of micromobility crashes that occurred in the City of Miami Beach and Downtown Miami areas, respectively.

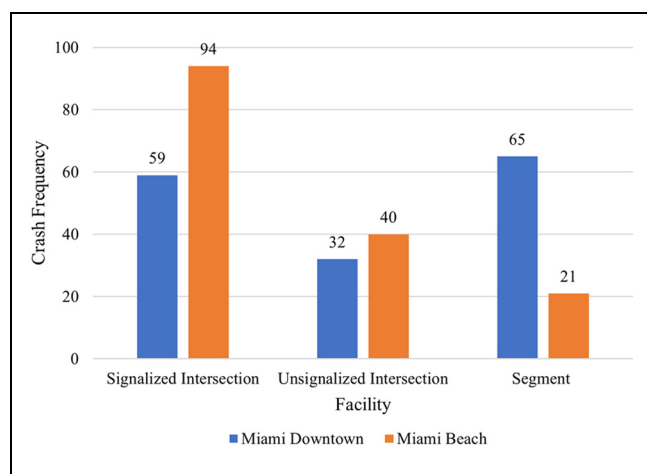
The proportion of roadway segments with zero recorded crashes was 77.4%, whereas the proportions for four-legged and three-legged signalized intersections

Table 2. Descriptive Statistics of Signalized Intersections

Attribute	Attribute category	Three-leg intersection	Four-leg intersection
Number of sites	na	113	262
Crash frequency ^a	na	32	114
AADT on major roads (vpd)	Minimum value	2,050	2,050
	Maximum value	82,000	121,000
	Average	20,708	20,849
	SD	10,997	13,852
No. of legs	na	3	4
No. of approaches with left-turn lanes	0	57	66
	1	42	61
	2	14	81
	3	NA	28
	4	NA	26
No. of approaches with right-turn lanes	0	79	154
	1	31	77
	2	4	30
	3	NA	1
No. of approaches with permitted left-turn signal	0	18	56
	1	21	50
	2	54	74
	3	20	18
	4	NA	66
No. of approaches with protected-permitted left-turn signal	0	77	107
	1	17	48
	2	12	64
	3	7	18
	4	NA	25
No. of approaches with no turn on red	0	102	226
	1	11	25
	2	NA	5
	3	NA	2
	4	NA	4
Presence of bike lane	No	37	49
	Yes	76	213
Presence of lighting	No	NA	NA
	Yes	113	262

Note: AADT = annual average daily traffic; SD = standard deviation; vpd = vehicles per day; na = not applicable; NA = not available.

^aCrashes involving micromobility users.

**Figure 2.** Crash statistics by facility type.

with zero recorded crashes were 70.5% and 77.4%, respectively.

Methodology

A cross-sectional analysis was utilized to estimate CMFs for micromobility crashes in Florida. Cross-sectional studies are effective for estimating CMFs when before-and-after studies are not feasible. This limitation might arise from insufficient crash data being available before and after implementing a specific engineering measure, uncertainty about when treatment was applied, or difficulties in isolating the impact of a countermeasure from other influencing factors (10). These challenges are particularly relevant when estimating CMFs for micromobility crashes in Florida. In cross-sectional studies, crash

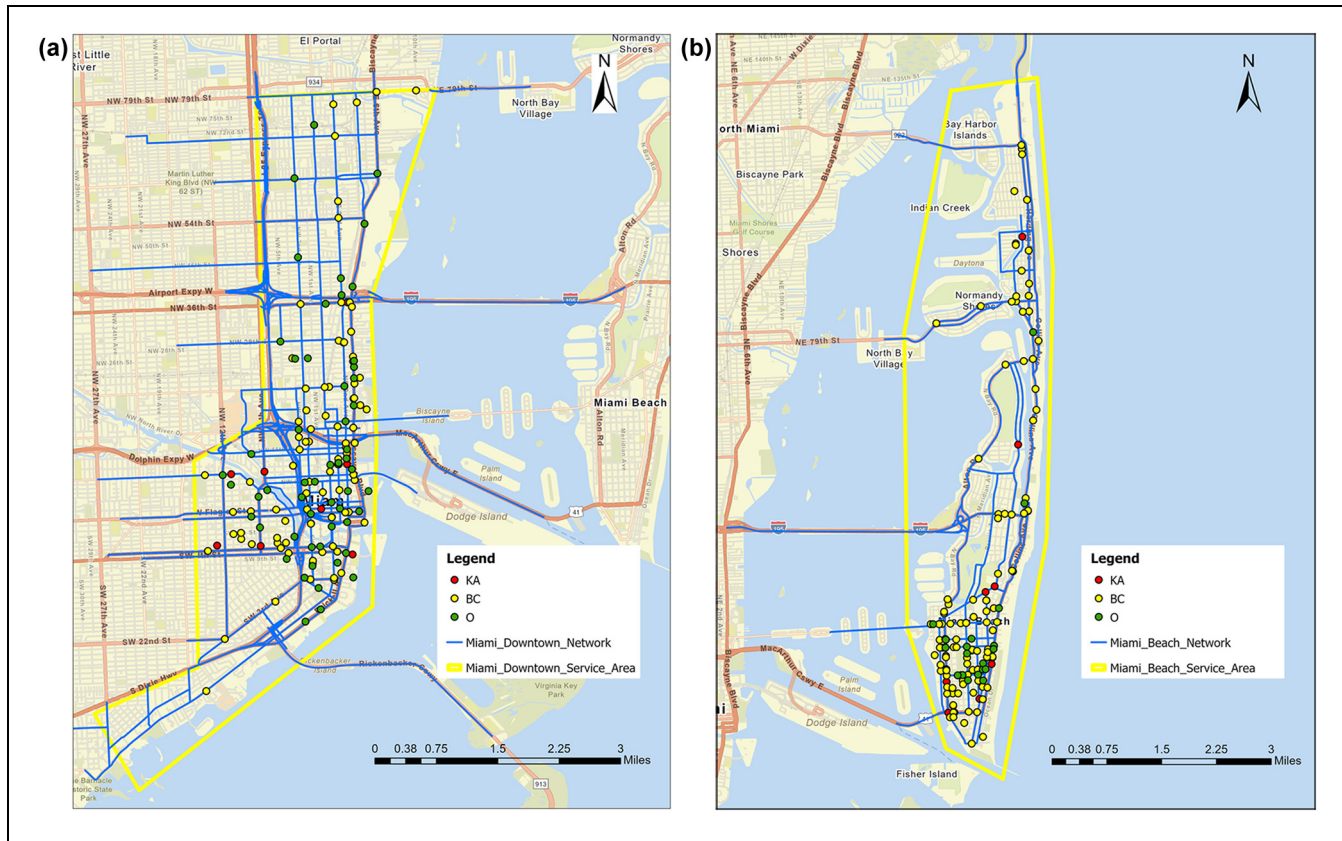


Figure 3. Spatial distribution of micromobility crashes in (a) Downtown Miami and (b) City of Miami Beach.

Note: K = fatal; A = incapacitating injury; B = nonincapacitating injury; C = possible injury; O = none.

experience at locations with and without a specific feature is studied, and then the difference in safety is attributed to that feature.

This research used an NB-L model to develop the relevant multivariate regression models. In these models, the dependent variable, or response variable, is the crash frequency, whereas the independent variables, or explanatory variables, are the roadway characteristics.

Multivariate regression models attempt to address all the variables that influence safety (44). The models are developed using crash data from sites both with and without the treatment/countermeasure. The change in crashes from a unit change in a specific variable can be estimated from the regression model (10).

The NB model is favored over the Poisson model because it accounts for overdispersion in crash data. It can be seen as a sequence of Bernoulli trials or a mixture of Poisson and gamma distributions. The formula for the NB GLM is represented by Equation 2 (28),

$$P(Y = y; \mu, \delta) = \text{NB}(y; \mu, \delta) = \frac{\Gamma(\delta + y)}{\Gamma(\delta)\Gamma(y + 1)} \left(\frac{\delta}{\mu + \delta} \right)^\delta \left(\frac{\mu}{\mu + \delta} \right)^y; \delta > 0, \mu > 0 \quad (2)$$

where μ is the mean response of the observations and δ indicates inverse of the dispersion parameter.

To address excess zeros, the NB-L model was introduced (32). This combines the NB distribution and the one-parameter Lindley distribution, resulting in a more flexible structure and greater degrees of freedom than the traditional NB distribution. The Lindley distribution enhances the NB model's ability to manage excess zeros (45). The Lindley distribution consists of a combination of two gamma distributions as represented by Equation 3 (30), which can be restructured as Equation 4 (22) as follows:

$$\eta \sim \frac{1}{1 + \gamma} \text{Gamma}(2, \gamma) + \left(1 - \frac{1}{1 + \gamma} \right) \text{Gamma}(1, \gamma) \quad (3)$$

$$\eta \sim \sum \text{Gamma}(1 + z, \gamma) \text{Bernoulli} \left(z; \frac{1}{1 + \gamma} \right) \quad (4)$$

where z is the Bernoulli random variable and γ is the rate parameter of the Lindley distribution.

Let $Y_i \sim \text{NB}$ distribution with shape parameter, p_i , and overdispersion parameter, δ . The hierarchical expression for the NB-L GLM is defined in Equation 5 (28, 32),

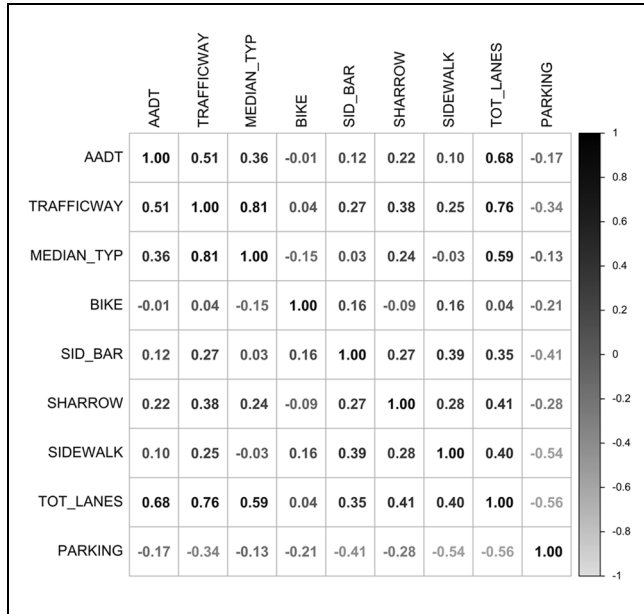


Figure 4. Variable correlation for roadway segment model.

Note: AADT = section average annual daily traffic; TOT_LANES = total number of lanes; MEDIAN_TYP = median type; BIKE = presence of bike lane; SHARROW = presence of sharrows; SIDEWALK = presence of sidewalk; SID_BAR = presence of sidewalk barrier; TRAFFICWAY = road type; PARKING = on-street parking.

$$Y_i \sim NB(p_i, \delta); \delta > 0, 0 < p_i < 1 \quad (5)$$

$$p_i = e^{-\eta_i}$$

$$\delta \sim \pi_\delta$$

$$\eta_i \sim \text{Lindley}(\gamma_i)$$

$$\gamma_i = e^{\beta X_i}$$

$$\beta \sim \pi_\beta$$

where γ is Lindley parameter; $X_i = (1, X_1, X_2, \dots, X_m)$, $\beta = (\beta_0, \beta_1, \dots, \beta_m)$, the vector of coefficients to be estimated; and π_δ and π_β the prior distribution for δ and β , respectively.

Specifying prior distributions for parameters is crucial for obtaining Bayesian estimates, as they reflect the prior knowledge. The site-specific frailty term uses a noninformative prior based on a gamma distribution (30). A weakly informative prior may lead to a model where the Lindley distribution's parameter estimate contributes more than that of the NB distribution. The model estimation involved four Markov chains with 10,000 iterations each. The analysis also included an offset that equaled the natural logarithm of three times the segment length for segments to predict crash frequency in crashes per mile. The number 3 was used since the analysis period was 3 years.

CMFs can be inferred from the estimated model parameters (46), that is, coefficients; and as the model form is log-linear, the CMFs can be calculated as the exponent of the associated coefficient of the countermeasure variable (47), as shown in Equation 6,

$$\text{CMF} = \exp(\beta_k \times (X_{kt} - X_{ku})) = \exp(\beta_k) \quad (6)$$

where X_{kt} represents roadway characteristic k (i.e., countermeasure) of a treated site, and X_{ku} is roadway characteristic k (i.e., countermeasure) of an untreated site.

Results and Discussion

This section discusses the estimated CMFs for micromobility crashes for the following three facility types: roadway segments, four-leg signalized intersections, and three-leg signalized intersections. An NB-L model for the roadway segments was developed. The following nine variables were considered in the model development: section AADT, number of lanes, median type, presence of bike lane, presence of sharrows, presence of sidewalk, presence of sidewalk barrier, trafficway, and on-street parking.

Separate NB-L models were developed to estimate CMFs at four-leg and three-leg signalized intersections. The following eight variables were considered in the model development: major road AADT, number of approaches with right-turn lane, number of approaches with left-turn lane, presence of bike lane, presence of lighting, number of approaches with permitted left-turn signal, number of approaches with protected-permitted left-turn signal, and number of approaches with no turn on red.

Roadway Segments

This section discusses the estimated CMFs for micromobility-related crashes on roadway segments. Before fitting the model, the correlation between variables was examined. This step is crucial because including multiple highly correlated variables can be superfluous or even impair the model's effectiveness. Figure 4 presents the variable correlation for the roadway segment model. It is worth noting that the selection of levels within the variables affects the correlation values. The cutoff point selected for this study was 0.70 (48–50). Based on the results from Figure 4, the variables "Trafficway" and "Median type" were highly correlated, and the latter was used in the model formulation. Also, the variable "Number of lanes" was removed from the model as it was correlated to variables "AADT," "Median type," and "Trafficway."

Table 3 presents the results of the NB-L model for roadway segments data. Table 4 presents the summary

Table 3. Negative Binomial Lindley Model Results for Roadway Segments

Base	Level	Estimate	SE	BCI	
				Lower limit	Upper limit
In (Section AADT)		0.262	0.735	−1.150	1.725
Median type	No median*				
	Paved	1.104	0.774	0.004 ^a	2.222 ^a
	Curb and vegetation	−0.600	0.712	−2.030	0.773
Presence of sidewalk barrier	No*				
	Yes	−1.410	0.670	−2.826 ^b	−0.185 ^b
Presence of bike lane	No*				
	Yes	−0.616	0.652	−1.905	0.651
Presence of on-street parking	No*				
	One sided	0.319	1.081	−1.826	2.429
	Two sided	1.899	0.829	0.360 ^b	3.617 ^b
Presence of sharrow lane	No*				
	Yes	1.470	0.689	0.166 ^b	2.884 ^b
Presence of sidewalk	No*				
	Yes	−0.764	0.836	−2.418	0.887

Note: BCI = Bayesian credible interval; AADT = annual average daily traffic; SE = standard error.

Values in bold are significant at a 90% BCI or higher.

Significant at: ^a85% BCI; ^b95% BCI.

*Indicates base condition.

Table 4. CMFs for Micromobility Crashes on Roadway Segments

Variable	Estimate	CMFs
Median type: Paved ^a	1.071	3.0 ¹
Median type: Curb and vegetation	0.313	NS
Section AADT	1.144	NS
Presence of sidewalk barrier ^b	−1.512	0.2 ²
Presence of bike lane	−0.597	NS
On-street parking: one sided	−2.558	NS
On-street parking: two sided ^c	1.865	6.7 ²
Presence of sharrow lane ^d	1.261	4.4 ²
Presence of sidewalk	−1.057	NS

Note: CMF = crash modification factor; AADT = annual average daily traffic; NS = not significant.

^aBase condition for type of median is no median.

^bBase condition for sidewalk barrier is absence of a sidewalk barrier.

^cBase condition for on-street parking is absence of on-street parking.

^dBase condition for sharrow lane is absence of sharrow lane.

Significant at: ¹ 85% BCI; ² 95% BCI.

of the estimated CMFs for total crashes involving micromobility devices on roadway segments. Table 4 includes the model coefficients and the corresponding CMFs. The variables that were significant at a 90% Bayesian credible interval (BCI) were used to estimate the CMFs.

As shown in Table 3, the following three variables were found to be significant at a 90% BCI: presence of a sidewalk barrier, presence of on-street parking, and presence of a sharrow lane. The CMF for the presence of on-street parking on both sides of the roadway was estimated at 6.7. This indicates that two-sided on-street

parking increased the probability of micromobility-related crashes by almost 6.7 times compared with roadways with no on-street parking. The presence of a sidewalk barrier had an estimated CMF of 0.2, indicating a reduction in micromobility crashes by 80%. This observation can be attributed to sidewalk barriers preventing riders from unexpectedly entering the roadway, especially when they are riding on sidewalks, and likewise, preventing vehicles from easily encroaching on the sidewalk. Finally, the presence of a sharrow lane, with a CMF of 4.4, showed an increase in the probability of micromobility crashes by almost 4.4 times. This observation can be attributed to how, in sharrow lanes, micromobility riders are mixed with the normal traffic—considering the speed differential, such circumstances are not safe for vulnerable micromobility riders.

On the other hand, the variable median type was found to be significant at 85% BCI. The estimated CMF for the paved median type was 3.0. This indicated that paved medians increased the probability of micromobility-related crashes by almost 3.0 times compared with roadways with no median.

The following is the summary of the estimated CMFs for micromobility crashes on roadway segments:

- The presence of on-street parking increases the probability of micromobility-related crashes by 6.7 times compared with roadway segments without on-street parking.
- The presence of a sidewalk barrier of any kind decreases the probability of micromobility-related

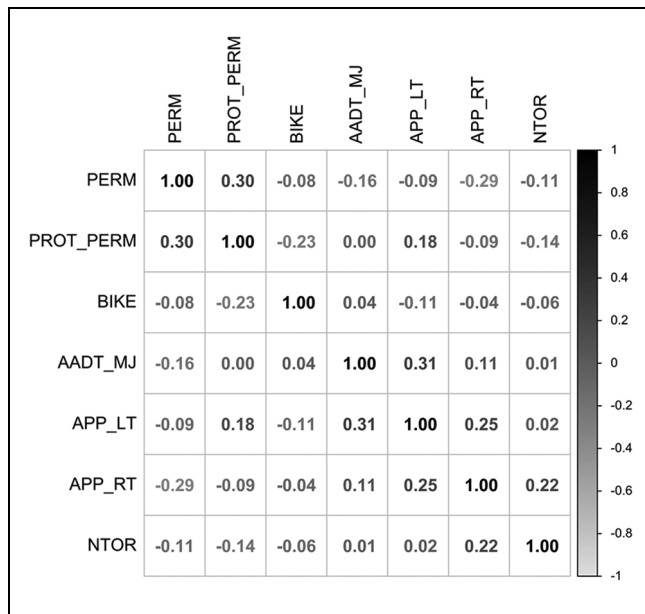


Figure 5. Variable correlation for four-leg signalized intersection model.

Note. AADT_MJ = major road average annual daily traffic;

APP_RT = number of approaches with an exclusive right-turn lane;

APP_LT = number of approaches with an exclusive left-turn lane;

BIKE = presence of bike facility; PERM = number of approaches with

permitted left-turn signal; PROT_PERM = number of approaches with

protected-permitted left-turn signal; NTOR = number of approaches with no turn on red sign.

crashes by 80% compared with when the roadway segment does not have a sidewalk barrier.

- The presence of a sharrows lane increases the probability of micromobility-related crashes by over 4 times compared with when the roadway segment does not have a bike lane.

Four-Leg Signalized Intersections

This section discusses the estimated CMFs for micromobility-related crashes on four-leg signalized intersections. To ensure a robust analysis, a correlation analysis between variables was conducted. No variables were highly correlated beyond a cutoff point of 0.70 (48–50). Figure 5 presents the variable correlation for the four-leg signalized intersections model.

Table 5 presents the results of the NB-L model for four-leg signalized intersections. Table 6 summarizes the estimated CMFs for crashes involving micromobility devices on four-leg signalized intersections. Table 6 includes the model coefficients and the corresponding CMFs. The variables that were significant at a 90% BCI were used to estimate CMFs.

As shown in Table 5, the following three variables were found to be significant at 90% BCI: the number of

approaches with a left-turn lane, the number of approaches with a permitted left-turn signal, and the number of approaches with a protected-permitted left-turn signal.

Intersections with three approaches with exclusive left-turn lanes had a CMF of 2.6, indicating that left-turn lanes at intersections were 160% more likely to increase the probability of micromobility crashes per year. This observation was expected since intersections with exclusive left-turn lanes indicate higher left-turning volumes, which in turn imply higher potential conflicts with micromobility riders. Intersections with one or four approaches with a protected-permitted left-turn signal phase had a CMF of 0.3 and 0.4, respectively, indicating a decrease in the probability of micromobility crashes by almost 70% and 60% per year. This occurred because such signal phasing allows left-turning vehicles to move separately from through traffic during the protected phase, thereby minimizing the chances of collisions with micromobility riders who are crossing the intersection. Previous studies have reached similar conclusions (44, 51). For permitted left-turns, when all approaches of a four-legged intersection have a permitted left-turn phase, the probability of micromobility crashes was shown to increase by over two times (CMF = 2.1). This observation may be explained by permitted left-turns being indicative of a low left-turning volume, which in turn indicates fewer conflicts with micromobility riders.

The following variables were found to be significant at 85%, 80%, or 75% BCI. The presence of a bike facility, the number of approaches with a left-turn lane, and the number of approaches with a permitted left-turn signal.

Based on Table 6, the following is the summary of the estimated CMFs for micromobility crashes at four-leg signalized intersections:

- Three approaches with exclusive left-turn lanes increase micromobility crashes by 2.6 times compared with intersections with no exclusive left-turn lane;
- Four approaches with permitted left-turn signals increase micromobility crashes by 2.1 times compared with intersections with no approaches with permitted left-turn signal;
- Four approaches with protected-permitted left-turn signals decrease micromobility crashes by 68% compared with intersections with no approaches with protected-permitted left-turn signals; and
- One approach with protected-permitted left-turn signals decreases micromobility crashes by 73% compared with intersections with no approaches with protected-permitted left-turn signals.

Table 5. Negative Binomial Lindley Model Results for Four-Leg Signalized Intersections

Base	Levels	Estimate	SE	BCI	
				Lower limit	Upper limit
In (Major road AADT)		0.306	0.461	−0.593	1.22
No. of approaches with Left-turn lanes	0 approach with exclusive left-turn lanes*				
	1 approach with exclusive left-turn lanes	0.493	0.396	0.037 ^a	0.950 ^a
	2 approaches with exclusive left-turn lanes	0.446	0.376	0.015 ^a	0.879 ^a
	3 approaches with exclusive left-turn lanes	0.951	0.472	0.021^b	1.879^b
	4 approaches with exclusive left-turn lanes	0.804	0.544	0.009 ^c	1.575 ^c
No. of approaches with right-turn lanes	0 approach with right-turn lanes*				
	1 approach with right-turn lanes	−0.184	0.323	−0.825	0.437
	2 approaches with right-turn lanes	−0.386	0.495	−1.375	0.555
	3 approaches with right-turn lanes	0.033	1.762	−3.546	3.514
Presence of a bike facility	No*				
	Yes	0.431	0.367	0.014 ^a	0.854 ^a
No. of approaches with permitted left-turn	0 approach with permitted left-turn*				
	1 approach with permitted left-turn	0.767	0.650	0.016 ^a	1.509 ^a
	2 approaches with permitted left-turn	0.752	0.531	0.072 ^d	1.425 ^d
	3 approaches with permitted left-turn	−0.653	0.831	−2.368	0.876
	4 approaches with permitted left-turn	−0.764	0.434	0.049^e	1.472^e
No. of approaches with protected-permitted left-turn	0 approach with protected-permitted left-turn*				
	1 approach with protected-permitted left-turn	−1.328	0.701	−2.494^e	−0.189^e
	2 approaches with protected-permitted left-turn	−0.297	0.520	−1.309	0.734
	3 approaches with protected-permitted left-turn	−0.142	0.683	−1.507	1.181
	4 approaches with protected-permitted left-turn	−0.962	0.531	−1.855^e	−0.111^e
No. of approaches with no turn on red	0 approach with no turn on red*				
	1 approach with no turn on red	0.112	0.433	−0.756	0.938
	2 approaches with no turn on red	−0.128	0.986	−2.204	1.694
	3 approaches with no turn on red	−3.461	3.326	−10.99	1.786
	4 approaches with no turn on red	−0.327	1.035	−1.843	2.250

Note: BCI = Bayesian credible interval; AADT = annual average daily traffic; SE = standard error.

Values in bold are significant at 90% BCI or higher.

Significant at: ^a 75% BCI; ^b 95% BCI; ^c 85% BCI; ^d 80% BCI; ^e 90% BCI.

*Indicates base condition.

Table 6. CMFs for Micromobility Crashes at Four-Leg Signalized Intersections

Variables	Estimate	CMFs
Major Road AADT	0.306	NS
1 approach with left-turn lanes ^a	0.493	1.6 ¹
2 approaches with left-turn lanes	0.446	1.6 ¹
3 approaches with left-turn lanes	0.951	2.6 ²
4 approaches with left-turn lanes	0.804	2.2 ³
1 approach with right-turn lanes	−0.184	NS
2 approaches with right-turn lanes	−0.386	NS
3 approaches with right-turn lanes	0.033	NS
Presence of bike facility ^b	0.431	1.5 ¹
1 approach with permitted left-turn ^c	0.767	2.2 ¹
2 approaches with permitted left-turn	0.752	2.1 ⁴
3 approaches with permitted left-turn	−0.653	NS
4 approaches with permitted left-turn	−0.764	2.1 ⁵
1 approach with protected-permitted left-turn ^d	−1.328	0.3 ⁵
2 approaches with protected-permitted left-turn	−0.297	NS
3 approaches with protected-permitted left-turn	−0.142	NS
4 approaches with protected-permitted left-turn	−0.962	0.4 ⁵
1 approach with no turn on red	0.112	NS
2 approaches with no turn on red	−0.128	NS
3 approaches with no turn on red	−3.461	NS
4 approaches with no turn on red	−0.327	NS

Note: CMF = crash modification factor; AADT = annual average daily traffic; BCI = Bayesian credible interval; NS = not significant.

^aBase condition for number of approaches with left-turn lanes is 0 approaches with left-turn lanes.

^bBase condition for presence of bike facility is absence of bike facility.

^cBase condition for number of approaches with permitted left-turn is 0 approaches with permitted left-turn.

^dBase condition for number of approaches with protected-permitted left-turn is 0 approaches with protected-permitted left-turn.

Significant at: ¹ 75% BCI; ² 95% BCI; ³ 85% BCI; ⁴ 80% BCI; ⁵ 90% BCI.

Three-Leg Signalized Intersections

This section discusses the estimated CMFs for micromobility-related crashes on three-leg signalized intersections. To ensure a robust analysis, the lighting condition variable was removed from the analysis since it only had one level. A correlation analysis between variables was also conducted. No variables were highly correlated beyond a cutoff point of 0.70 (48–50). Figure 6 presents the variable correlation for the three-leg signalized intersection model.

Table 7 presents the NB-L model results for three-leg signalized intersections. Table 8 presents the summary of the estimated CMFs for crashes involving micromobility devices on three-leg signalized intersections. It includes the model coefficients and the corresponding CMFs. The variables that were significant at a 90% BCI were used to estimate the CMFs.

As shown in Table 7, two variables were found to be significant at a 90% BCI: the number of approaches with a protected-permitted left-turn signal and the number of approaches with exclusive right-turn lanes. Intersections with three approaches with protected-permitted left-turn signal phases had a CMF of 0.01, indicating a decrease in the probability of micromobility crashes by about 99% compared with intersections with no approaches with

protected-permitted left-turn signal phases. This may be attributed to the reduced conflict points between left-turning vehicles and micromobility users at these intersections during protected signal phases. Intersections with one approach with an exclusive right-turn lane had a CMF of 2.7, indicating a 170% increase in crashes compared with three-legged intersections with no approaches with right-turn lanes.

Based on the results from Table 8, the following is a summary of the estimated CMFs for micromobility crashes at three-leg signalized intersections.

- One approach with an exclusive right-turn lane increases micromobility crashes by 2.7 times compared with intersections with no approaches with an exclusive right-turn lane; and
- Three approaches with a protected-permitted left-turn signal decrease micromobility crashes by 99% compared with intersections with no approach with a protected-permitted left-turn signal.

In summary, Tables 9 and 10 present the CMFs and corresponding CRFs estimated for roadway segments and signalized intersections, respectively.

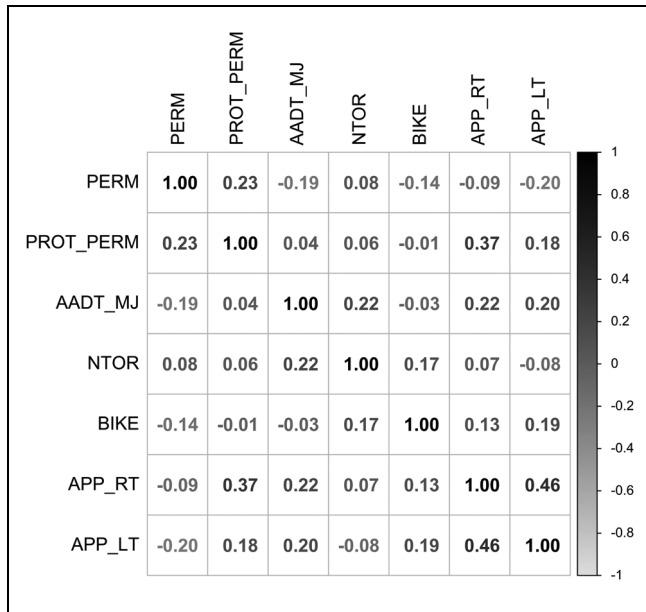


Figure 6. Variable correlation for three-leg signalized intersection model.

Note. AADT_MJ = major road average annual daily traffic; APP_RT = number of approaches with an exclusive right-turn lane; APP_LT = number of approaches with an exclusive left-turn lane; BIKE = presence of bicycle facility; PERM = number of approaches with permitted left-turn signal; PROT_PERM = number of approaches with protected-permitted left-turn signal; NTOR = number of approaches with no turn on red sign.

Conclusions and Recommendations

The objective of this study was to estimate Florida-specific CMFs for micromobility crashes at roadway segments and signalized intersections. Micromobility devices, which include small, lightweight vehicles such as e-scooters and e-bikes, have seen a significant rise in usage across the United States. As these systems become more prevalent, it is paramount to identify which infrastructure or geometric characteristics of roadways favor micromobility and improve safety. Most agencies have designated service areas where micromobility vehicles are allowed to operate. Therefore, understanding and addressing the specific needs of these road users is crucial for the development of safe and efficient transportation networks.

Estimating CMFs for various roadway features assists in decision making and conducting cost-assessment analyses for new roads and improvements of existing ones in relation to crash reduction. These CMFs provide valuable insights for transportation planners and engineers to design safer roadways and implement effective safety measures. By understanding the impact of different roadway characteristics on micromobility crashes, targeted interventions can be made to enhance overall traffic safety.

The developed CMFs were based on crash data from the Downtown Miami and City of Miami Beach areas

Table 7. Negative Binomial Lindley Model Results for Three-Leg Signalized Intersections

Base	Level	Estimate	SE	BCI	
				Lower limit	Upper limit
In (Major road AADT)		-0.550	0.750	-2.010	0.935
No. of approaches with left-turn lanes	0 approach with left-turn lanes*				
	1 approach with left-turn lanes	-0.107	0.526	-0.933	1.128
	2 approaches with left-turn lanes	-0.950	0.906	-2.804	0.763
No. of approaches with right-turn lanes	0 approach with right-turn lanes*				
	1 approach with right-turn lanes	0.986	0.579	0.045^a	1.941^a
	2 approaches with right-turn lanes	-0.322	1.429	-3.476	2.154
Presence of bicycle facility	No*				
	Yes	0.118	0.518	-0.892	1.145
No. of approaches with permitted left-turn	0 approach with permitted left-turn*				
	1 approach with permitted left-turn	1.055	0.917	0.041 ^b	2.082 ^b
	2 approaches with permitted left-turn	0.219	0.692	-1.093	1.633
	3 approaches with permitted left-turn	0.345	0.815	-1.268	1.942

(continued)

Table 7. (continued)

Base	Level	Estimate	SE	BCI	
				Lower limit	Upper limit
No. of approaches with protected-permitted left-turn	0 approach with protected-permitted left-turn*				
	1 approach with protected-permitted left-turn	−0.973	0.814	−1.915 ^b	0.047 ^b
	2 approaches with protected-permitted left-turn	−0.257	0.761	−1.812	1.178
	3 approaches with protected-permitted left-turn	−4.993	2.923	−11.72^c	−0.503^c
No. of approaches with no turn on red	0 approaches with no turn on red*				
	1 approaches with no turn on red	0.023	0.763	−1.533	1.487

Note: BCI = Bayesian credible interval; AADT = annual average daily traffic; SE = standard error.

Values in bold are significant at 90% BCI or higher.

Significant at: ^a 90% BCI; ^b 75% BCI; ^c 95% BCI.

*Indicates base condition.

Table 8. CMFs for Micromobility Crashes at Three-Leg Signalized Intersections

Variables	Estimate	CMFs
Major road AADT	−0.550	NS
1 approach with left-turn lanes	−0.107	NS
2 approaches with left-turn lanes	−0.950	NS
1 approach with right-turn lanes ^a	0.986	2.7 ¹
2 approaches with right-turn lanes	−0.322	NS
Presence of bicycle facility	0.118	NS
1 approach with permitted left-turn ^b	1.055	2.9 ²
2 approaches with permitted left-turn	0.219	NS
3 approaches with permitted left-turn	0.345	NS
1 approach with protected-permitted left-turn ^c	−0.973	0.4 ²
2 approaches with protected-permitted left-turn	−0.257	NS
3 approaches with protected-permitted left-turn	−4.993	0.01 ³
1 approach with no turn on red	0.023	NS

Note: CMF = crash modification factor; AADT = annual average daily traffic; BCI = Bayesian credible interval; NS = not significant.

^aBase condition for number of approaches with exclusive right-turn is 0 approaches with exclusive right-turn.

^bBase condition for number of approaches with permitted left-turn is 0 approaches with permitted left-turn.

^cBase condition for number of approaches with protected-permitted left-turn is 0 approaches with protected-permitted left-turn.

Significant at: ¹ 90% BCI; ² 75% BCI; ³ 95% BCI.

for the years 2021 to 2023. An extensive review of police reports was conducted using a text-mining approach to identify crashes involving micromobility riders, as there

is currently no specific description to filter out micromobility-related crashes. The other data utilized in this study included roadway characteristics such as section AADT, number of lanes, median type, presence of bicycle lanes, presence of sharrows, presence of sidewalks, presence of sidewalk barriers, trafficway, and presence of on-street parking for roadway segments. For signalized intersections, the variables included major road AADT, number of approaches with a right-turn lane, number of approaches with a left-turn lane, presence of bicycle lanes, presence of lighting, number of approaches with permitted left-turn signals, number of approaches with protected-permitted left-turn signals, and number of approaches with no turn on red. Most of the variables, except for AADT values, were obtained manually from Google Maps.

The NB-L model was developed to estimate CMFs. The results from the NB-L model for roadway segments indicated that three variables, presence of a sidewalk barrier, presence of a sharrows lane, and presence of on-street parking were significant at a 90% BCI. The presence of on-street parking (CMF = 6.7) and the presence of sharrows lanes (CMF = 4.4) were found to increase the probability of micromobility-related crashes, whereas the presence of a sidewalk barrier (CMF = 0.2) was associated with a decrease in the probability of micromobility crashes.

The results from the NB-L model for four-legged intersections showed three significant variables at a 90% BCI: number of approaches with left-turn lanes, number of approaches with permitted left-turn signals, and number of approaches with protected-permitted left-turn

Table 9. Summary of the Estimated CMFs and CRFs for Roadway Segments

Base condition	Level	CMF	CRF, %
Absence of sidewalk barrier	Presence of sidewalk barrier	0.2	80
Absence of sharrow lane	Presence of sharrow lane	4.4	−340
Absence of on-street parking	Two-sided on-street parking	6.7	−570

Note: CMF = crash modification factor; CRF = crash reduction factor; BCI = Bayesian credible interval.
Values presented are significant at a 90% BCI or higher.

Table 10. Summary of the Estimated CMFs and CRFs for Signalized Intersections

Base condition	Level	Four-leg intersection		Three-leg intersection	
		CMF	CRF, %	CMFs	CRF, %
No approaches with left-turn lanes	3 approaches with left-turn lanes	2.6	−160	NS	NS
No approaches with right-turn lanes	1 approach with right-turn lane	NS	NS	2.7	−170
No approaches with permitted left-turn signal	4 approaches with permitted left-turn signal	2.1	−110	NS	NS
No approaches with protected-permitted left-turn signal	1 approach with protected-permitted left-turn signal	0.3	70	NS	NS
	3 approaches with protected-permitted left-turn signal	NS	NS	0.01	99
	4 approaches with protected-permitted left-turn signal	0.4	60	NS	NS

Note: CMF = crash modification factor; CRF = crash reduction factor; BCI = Bayesian credible interval; NS = not significant.
Values presented are significant at a 90% BCI or higher.

signals. Intersections with three approaches with exclusive left-turn lanes had a CMF of 2.6, indicating a 2.6 times increase in the probability of micromobility crashes. Intersections with one and four approaches with a protected-permitted left-turn signal had CMFs of 0.3 and 0.4, respectively, reducing crash probability. Four approaches with permitted left-turn phases increased the micromobility crash probability (CMF = 2.1).

The results from the NB-L model for three-leg intersections showed that two variables were significant at a 90% BCI. Three-leg intersections with all three approaches with protected-permitted left-turn signals were associated with a 99% decrease in the probability of micromobility crashes (CMF = 0.01). In contrast, intersections with one approach featuring an exclusive right-turn lane were associated with an increased probability of micromobility crashes (CMF = 2.7).

Cities and agencies planning to deploy micromobility systems can greatly benefit from these findings, which offer practical recommendations for improving micromobility safety on both roadway segments and intersections. Providing sidewalk barriers of any kind is essential to reducing crashes by separating micromobility users from motorized traffic. Sharrows lanes, however, should be avoided, as mixing riders with

motorized traffic significantly increases crash risk primarily because of the speed differential between micromobility devices and motor vehicles. For streets with medians, paved medians are not safer for micromobility users. At intersections, protected-permitted left-turn signals are preferable to permitted-only left-turn signals. Finally, intersection design should include better bike lane facilities, focusing on how micromobility users interact with motorized traffic, such as right-turning vehicles.

There are several limitations to this study, primarily related to data constraints. First, the study only considered crashes from the City of Miami Beach and Downtown Miami service areas. Since these cities are tourist destinations, the estimated CMFs may be more applicable to areas with similar characteristics. Second, the study relied on crashes reported to the police. It is reasonable to assume that many crashes involving micromobility riders go unreported. Additionally, accurately quantifying micromobility exposure, which is crucial for developing robust CPMs, remains challenging. However, the roadway segments included in our analysis were carefully selected based on designated micromobility service areas, zones where micromobility users are permitted and even encouraged to operate. This selection criterion

ensured that the analyzed segments are, at minimum, areas of potential micromobility activity.

Acknowledgments

The authors thank the Florida Department of Transportation for providing access to data utilized in this research.

Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: E. Mwambeleko, P. Alluri, T. Sando H.M. Nayem; data collection: E. Mwambeleko, H.M. Nayem, A. Aida, J. Matyenyi; analysis and interpretation of results E. Mwambeleko, P. Alluri, T. Sando, H.M. Nayem; draft manuscript preparation: E. Mwambeleko, P. Alluri, T. Sando, H.M. Nayem, A. Aida, J. Matyenyi. All authors reviewed the results and approved the final version of the manuscript.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This research was partially supported by the Florida Department of Transportation.

ORCID iDs

Enock Mwambeleko  <https://orcid.org/0000-0001-5792-2346>
Amy Aida  <https://orcid.org/0009-0009-3408-3119>

References

1. Shared Micromobility in 2023 | National Association of City Transportation Officials. <https://nacto.org/publication/shared-micromobility-in-2023/>. Accessed July 27, 2024.
2. Hosseinzadeh, A., M. Algomaiah, R. Kluger, and Z. Li. Spatial Analysis of Shared E-Scooter Trips. *Journal of Transport Geography*, Vol. 92, 2021, p. 103016. <https://doi.org/10.1016/j.jtrangeo.2021.103016>
3. Tian, D., A. D. Ryan, C. M. Craig, K. Sievert, and N. L. Morris. Characteristics and Risk Factors for Electric Scooter-Related Crashes and Injury Crashes among Scooter Riders: A Two-Phase Survey Study. *International Journal of Environmental Research and Public Health*, Vol. 19, No. 16, 2022, p. 10129. <https://doi.org/10.3390/ijerph191610129>
4. Oeschger, G., P. Carroll, and B. Caulfield. Micromobility and Public Transport Integration: The Current State of Knowledge. *Transportation Research Part D: Transport and Environment*, Vol. 89, 2020, p. 102628. <https://doi.org/10.1016/j.trd.2020.102628>
5. Abduljabbar, R. L., S. Liyanage, and H. Dia. The Role of Micro-Mobility in Shaping Sustainable Cities: A Systematic Literature Review. *Transportation Research Part D: Transport and Environment*, Vol. 92, 2021, p. 102734. <https://doi.org/10.1016/j.trd.2021.102734>
6. Pedestrian and Bicycle Information Center. *The Basics of Micromobility and Related Motorized Devices for Personal Transport*. Pedestrian and Bicycle Information Center, Chapel Hill, NC, 2019. https://www.pedbikeinfo.org/downloads/PBIC_Brief_MicromobilityTypology.pdf.
7. Shin, S., and S. Choo. Influence of Built Environment on Micromobility–Pedestrian Accidents. *Sustainability*, Vol. 15, No. 1, 2023, pp. 1–11. <https://doi.org/10.3390/su15010582>
8. Pourfalamatoun, S., J. Ahmed, and E. E. Miller. Shared Electric Scooter Users and Non-Users: Perceptions on Safety, Adoption and Risk. *Sustainability*, Vol. 15, No. 11, 2023, p. 9045. <https://doi.org/10.3390/su15119045>
9. Karpinski, E., E. Bayles, and T. Sanders. Safety Analysis for Micromobility: Recommendations on Risk Metrics and Data Collection. *Transportation Research Record: Journal of the Transportation Research Board*, 2022. 2676: 420–435.
10. Park, J., M. Abdel-Aty, J. Lee, and C. Lee. Developing Crash Modification Functions to Assess Safety Effects of Adding Bike Lanes for Urban Arterials with Different Roadway and Socio-Economic Characteristics. *Accident Analysis and Prevention*, Vol. 74, 2015, pp. 179–191. <https://doi.org/10.1016/j.aap.2014.10.024>
11. Bahar, G., M. Maslah, R. Wolff, and P. Park. Desktop Reference for Crash Reduction Factors. No. FHWA-SA-07-015. Federal Highway Administration, Washington, D.C., 2007. <https://rosap.ntl.bts.gov/view/dot/16438>.
12. Gross, F., B. Persaud, and C. Lyon. A Guide to Developing Quality Crash Modification Factors. No. FHWA-SA-10-032. Federal Highway Administration, Washington, D.C., 2010. <https://rosap.ntl.bts.gov/view/dot/41025>
13. Carter, D., R. Srinivasan, F. Gross, and F. Council. Recommended Protocols for Developing Crash Modification Factors. No. NCHRP, 20-07(314). Transportation Research Board, Washington, D.C., 2012.
14. Lord, D., S. P. Washington, and J. N. Ivan. Poisson, Poisson-Gamma and Zero-Inflated Regression Models of Motor Vehicle Crashes: Balancing Statistical Fit and Theory. *Accident Analysis and Prevention*, Vol. 37, No. 1, 2005, pp. 35–46. <https://doi.org/10.1016/j.aap.2004.02.004>
15. Shankar, V., J. Milton, and F. Mannering. Modeling Accident Frequencies as Zero-Altered Probability Processes: An Empirical Inquiry. *Accident Analysis and Prevention*, Vol. 29, No. 6, 1997, pp. 829–837. [https://doi.org/10.1016/S0001-4575\(97\)00052-3](https://doi.org/10.1016/S0001-4575(97)00052-3)
16. Qin, X., J. N. Ivan, and N. Ravishanker. Selecting Exposure Measures in Crash Rate Prediction for Two-Lane Highway Segments. *Accident Analysis and Prevention*, Vol. 36, No. 2, 2004, pp. 183–191. [https://doi.org/10.1016/S0001-4575\(02\)00148-3](https://doi.org/10.1016/S0001-4575(02)00148-3)
17. Lord, D., S. Washington, and J. N. Ivan. Further Notes on the Application of Zero-Inflated Models in Highway

- Safety. *Accident Analysis and Prevention*, Vol. 39, No. 1, 2007, pp. 53–57. <https://doi.org/10.1016/j.aap.2006.06.004>
18. Kumara, S. S. P., and H. C. Chin. Modeling Accident Occurrence at Signalized Tee Intersections with Special Emphasis on Excess Zeros. *Traffic Injury Prevention*, Vol. 4, No. 1, 2003, pp. 53–57. <https://doi.org/10.1080/15389580309852>
19. Prasetijo, J., and W. Z. Musa. Modeling Zero—Inflated Regression of Road Accidents at Johor Federal Road F001. *MATEC Web of Conferences*, Vol. 47, 2016, pp. 1–7. <https://doi.org/10.1051/mateconf/20164703001>
20. Abdulhafedh, A. *Incorporating Zero-Inflated Poisson (ZIP) Regression Model in Crash Frequency Analysis*. Transportation Research Record, 2023. <https://doi.org/10.5281/zenodo.7632596>
21. Son, H. D., Y. J. Kweon, and B. B. Park. Development of Crash Prediction Models with Individual Vehicular Data. *Transportation Research Part C: Emerging Technologies*, Vol. 19, No. 6, 2011, pp. 1353–1363. <https://doi.org/10.1016/j.trc.2011.03.002>
22. Shaon, M. R. R., and X. Qin. Use of Mixed Distribution Generalized Linear Models to Quantify Safety Effects of Rural Roadway Features. *Transportation Research Record: Journal of the Transportation Research Board*, 2016. 2583: 134–141.
23. Shirazi, M., S. S. Dhavala, D. Lord, and S. R. Geedipally. A Methodology to Design Heuristics for Model Selection Based on the Characteristics of Data: Application to Investigate When the Negative Binomial Lindley (NB-L) Is Preferred over the Negative Binomial (NB). *Accident Analysis and Prevention*, Vol. 107, 2017, pp. 186–194. <https://doi.org/10.1016/j.aap.2017.07.002>
24. Lord, D., and S. R. Geedipally. The Negative Binomial-Lindley Distribution as a Tool for Analyzing Crash Data Characterized by a Large Amount of Zeros. *Accident Analysis and Prevention*, Vol. 43, No. 5, 2011, pp. 1738–1742. <https://doi.org/10.1016/j.aap.2011.04.004>
25. Zamani, H., and N. Ismail. *Negative Binomial-Lindley Distribution and Its Application* Hossein Zamani and Noriszura Ismail Actuarial Science Program. School of Mathematical Sciences, University Kebangsaan Malaysia, 43600 UKM Bangi, Selangor, Malaysia. Vol. 6, No. 1, 2010, pp. 4–9.
26. Khodadadi, A., I. Tsapakis, S. Das, D. Lord, and Y. Li. Application of Different Negative Binomial Parameterizations to Develop Safety Performance Functions for Non-Federal Aid System Roads. *Accident Analysis and Prevention*, Vol. 156, 2021, p. 106103. <https://doi.org/10.1016/j.aap.2021.106103>
27. Gil-Marin, J. K., M. Shirazi, and J. N. Ivan. Assessing the Negative Binomial-Lindley Model for Crash Hotspot Identification: Insights from Monte Carlo Simulation Analysis. *Accident Analysis and Prevention*, Vol. 199, 2024, p. 107478. <https://doi.org/10.1016/j.aap.2024.107478>
28. Geedipally, S. R., D. Lord, and S. S. Dhavala. The Negative Binomial-Lindley Generalized Linear Model: Characteristics and Application Using Crash Data. *Accident Analysis and Prevention*, Vol. 45, 2012, pp. 258–265. <https://doi.org/10.1016/j.aap.2011.07.012>
29. Anis, M., S. R. Geedipally, and D. Lord. *Pedestrian Crash Causation Analysis Near Bus Stops: Insights from Random Parameters NB-Lindley Models*. Accident Analysis & Prevention, Vol. 220, 2025, pp. 1–27.
30. Shaon, M. R. R., X. Qin, M. Shirazi, D. Lord, and S. R. Geedipally. Developing a Random Parameters Negative Binomial-Lindley Model to Analyze Highly over-Dispersed Crash Count Data. *Analytic Methods in Accident Research*, Vol. 18, 2018, pp. 33–44. <https://doi.org/10.1016/j.amar.2018.04.002>
31. Rusli, R., M. M. Haque, A. P. Afghari, and M. King. Applying a Random Parameters Negative Binomial Lindley Model to Examine Multi-Vehicle Crashes along Rural Mountainous Highways in Malaysia. *Accident Analysis and Prevention*, Vol. 119, 2018, pp. 80–90. <https://doi.org/10.1016/j.aap.2018.07.006>
32. Dzinyela, R., M. Shirazi, S. Das, and D. Lord. The Negative Binomial-Lindley Model with Time-Dependent Parameters: Accounting for Temporal Variations and Excess Zero Observations in Crash Data. *Accident Analysis and Prevention*, Vol. 207, 2024, p. 107711. <https://doi.org/10.1016/j.aap.2024.107711>
33. Tang, F., X. Fu, M. Cai, Y. Lu, S. Zhong, and C. Lu. Applying a Correlated Random Parameters Negative Binomial Lindley Model to Examine Crash Frequency along Highway Tunnels in China. *IEEE Access*, Vol. 8, 2020, pp. 213473–213488. <https://doi.org/10.1109/ACCESS.2020.3040211>
34. Islam, A. S. M. M., M. Shirazi, and D. Lord. Finite Mixture Negative Binomial-Lindley for Modeling Heterogeneous Crash Data with Many Zero Observations. *Accident Analysis and Prevention*, Vol. 175, 2022, p. 106765. <https://doi.org/10.1016/j.aap.2022.106765>
35. Islam, A. S. M. M., M. Shirazi, and D. Lord. Grouped Random Parameters Negative Binomial-Lindley for Accounting Unobserved Heterogeneity in Crash Data with Preponderant Zero Observations. *Analytic Methods in Accident Research*, Vol. 37, 2023, p. 100255. <https://doi.org/10.1016/j.amar.2022.100255>
36. Azimian, A., and J. Jiao. Modeling Factors Contributing to Dockless E-Scooter Injury Accidents in Austin, Texas. *Traffic Injury Prevention*, Vol. 23, No. 2, 2022, pp. 107–111. <https://doi.org/10.1080/15389588.2022.2030057>
37. Sikka, N., C. Vila, M. Stratton, M. Ghassemi, and A. Pourmand. Sharing the Sidewalk: A Case of E-Scooter Related Pedestrian Injury. *American Journal of Emergency Medicine*, Vol. 37, No. 9, 2019, pp. 1807.e5–1807.e7. <https://doi.org/10.1016/j.ajem.2019.06.017>
38. Maas, S., M. Attard, and M. A. Caruana. Assessing Spatial and Social Dimensions of Shared Bicycle Use in a Southern European Island Context: The Case of Las Palmas de Gran Canaria. *Transportation Research Part A: Policy and Practice*, Vol. 140, 2020, pp. 81–97. <https://doi.org/10.1016/j.tra.2020.08.003>
39. Fonseca-Cabrera, A. S., D. Llopis-Castelló, A. M. Pérez-Zuriaga, C. Alonso-Troyano, and A. Garcia. Micromobility Users' Behaviour and Perceived Risk during Meeting Manoeuvres. *International Journal of Environmental*

- Research and Public Health*, Vol. 18, No. 23, 2021, p. 12465. <https://doi.org/10.3390/ijerph182312465>
40. Cubells, J., C. Miralles-Guasch, and O. Marquet. E-Scooter and Bike-Share Route Choice and Detours: Modelling the Influence of Built Environment and Sociodemographic Factors. *Journal of Transport Geography*, Vol. 111, 2023, p. 103664. <https://doi.org/10.1016/j.jtrangeo.2023.103664>
 41. Kleinertz, H., A. Volk, D. Dalos, R. Rutkowski, K. H. Frosch, and D. M. Thiesen. Risk Factors and Injury Patterns of E-Scooter Associated Injuries in Germany. *Scientific Reports*, Vol. 13, No. 1, 2023, pp. 1–10. <https://doi.org/10.1038/s41598-022-25448-z>.
 42. Yang, H., Q. Ma, Z. Wang, Q. Cai, K. Xie, and D. Yang. Safety of Micro-Mobility: Analysis of E-Scooter Crashes by Mining News Reports. *Accident Analysis and Prevention*, Vol. 143, 2020, p. 105608. <https://doi.org/10.1016/j.aap.2020.105608>.
 43. Florida Department of Transportation. *Roadway Characteristics Inventory Handbook*. Transportation Data and Analytics Office, Florida Department of Transportation, Tallahassee, FL, 2024. https://ftp.fdot.gov/public/file/LA_R9yJQLUmsRqSrO2ZgEQ/FDOT_RCI_Handbook.pdf.
 44. Raihan, M. A., P. Alluri, W. Wu, and A. Gan. Estimation of Bicycle Crash Modification Factors (CMFs) on Urban Facilities Using Zero Inflated Negative Binomial Models. *Accident Analysis & Prevention*, Vol. 123, 2019, pp. 303–313. <https://doi.org/10.1016/j.aap.2018.12.009>
 45. Shirazi, M., D. Lord, and S. R. Geedipally. Sample-Size Guidelines for Recalibrating Crash Prediction Models: Recommendations for the Highway Safety Manual. *Accident Analysis & Prevention*, Vol. 93, 2016, pp. 160–168. <https://doi.org/10.1016/j.aap.2016.04.011>
 46. Al-Marafi, M. N., and K. Somasundaraswaran. A Review of the State-of-the-Art Methods in Estimating Crash Modification Factor (CMF). *Transportation Research Interdisciplinary Perspectives*, Vol. 20, 2023, p. 100839. <https://doi.org/10.1016/j.trip.2023.100839>
 47. Lord, D., and J. A. Bonneson. Development of Accident Modification Factors for Rural Frontage Road Segments in Texas. *Transportation Research Record: Journal of the Transportation Research Board*, 2007, 20–27.
 48. Champahom, T., S. Jomnonkwao, C. Banyong, W. Nambulee, A. Karoonsoontawong, and V. Ratanavaraha. Analysis of Crash Frequency and Crash Severity in Thailand: Hierarchical Structure Models Approach. *Sustainability*, Vol. 13, No. 18, 2021, p. 10086. <https://doi.org/10.3390/su131810086>
 49. Dormann, C. F., J. Elith, S. Bacher, C. Buchmann, G. Carl, G. Carré, J. R. G. Marquéz, et al. Collinearity: A Review of Methods to Deal with It and a Simulation Study Evaluating Their Performance. *Ecography*, Vol. 36, No. 1, 2013, pp. 27–46. <https://doi.org/10.1111/j.1600-0587.2012.07348.x>
 50. Quddus, M. A., C. Wang, and S. G. Ison. Road Traffic Congestion and Crash Severity: Econometric Analysis Using Ordered Response Models. *Journal of Transportation Engineering*, Vol. 136, No. 5, 2010, pp. 424–435. [https://doi.org/10.1061/\(ASCE\)TE.1943-5436.0000044](https://doi.org/10.1061/(ASCE)TE.1943-5436.0000044)
 51. Srinivasan, R., C. Lyon, B. Persaud, J. Baek, F. Gross, S. Smith, and C. Sundstrom. Crash Modification Factors for Changes to Left-Turn Phasing. *Transportation Research Record: Journal of the Transportation Research Board*, 2012, 2279: 108–117. <https://doi.org/10.3141/2279-13>

The opinions, findings, and conclusions or recommendations in this research paper are those of the authors and do not necessarily reflect the views of the Florida Department of Transportation or Florida International University or University of North Florida.